

Undetected and Undertreated

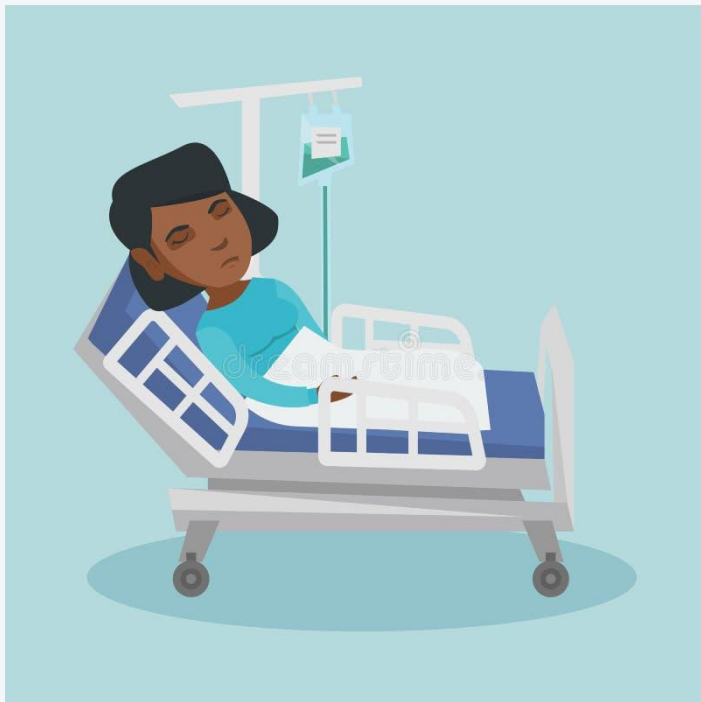
*Racial and Gender Disparities in Hidden Hypoxemia
and Their Economic Consequences*

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COMP/STAT 212 · Macalester College · Spring 2026

Sally's Case: Hidden Hypoxemia

Patients with high melanin are 3× more likely to have hidden hypoxemia



What is Hidden Hypoxemia?

Pulse oximeters clip to your fingertip and estimate blood oxygen saturation (SpO_2) using light absorption through the skin.

Hidden hypoxemia occurs when the device reads $\text{SpO}_2 \geq 88\%$ (appearing normal) while true arterial oxygen (SaO_2) is below 88% — the patient is **genuinely hypoxic but the device says otherwise**.

This causes clinicians to withhold oxygen therapy and delay critical interventions — from patients who need them.

WHY DARKER SKIN TONES?

Melanin absorption

Darker skin contains more melanin, absorbing the same near-infrared wavelengths the device uses — inflating the reading.

Device training bias

FDA-cleared oximeters were validated primarily on lighter-skinned subjects, building the bias into their calibration.

Clinical consequence

Black and Hispanic patients are significantly more likely to experience undetected hypoxemia, delaying treatment eligibility.

Two Complementary Datasets

DATASET 1 · OPENOXIMETRY

UCSF Hypoxia Lab

- Paired SpO₂ (pulse ox) and SaO₂ (arterial blood gas) measurements
- Fitzpatrick + Monk skin tone scores from spectrophotometry
- Self-reported race and ethnicity
- Key outcomes: bias (SpO₂ – SaO₂) and occult hypoxemia flag
- Access: Data Use Agreement via PhysioNet

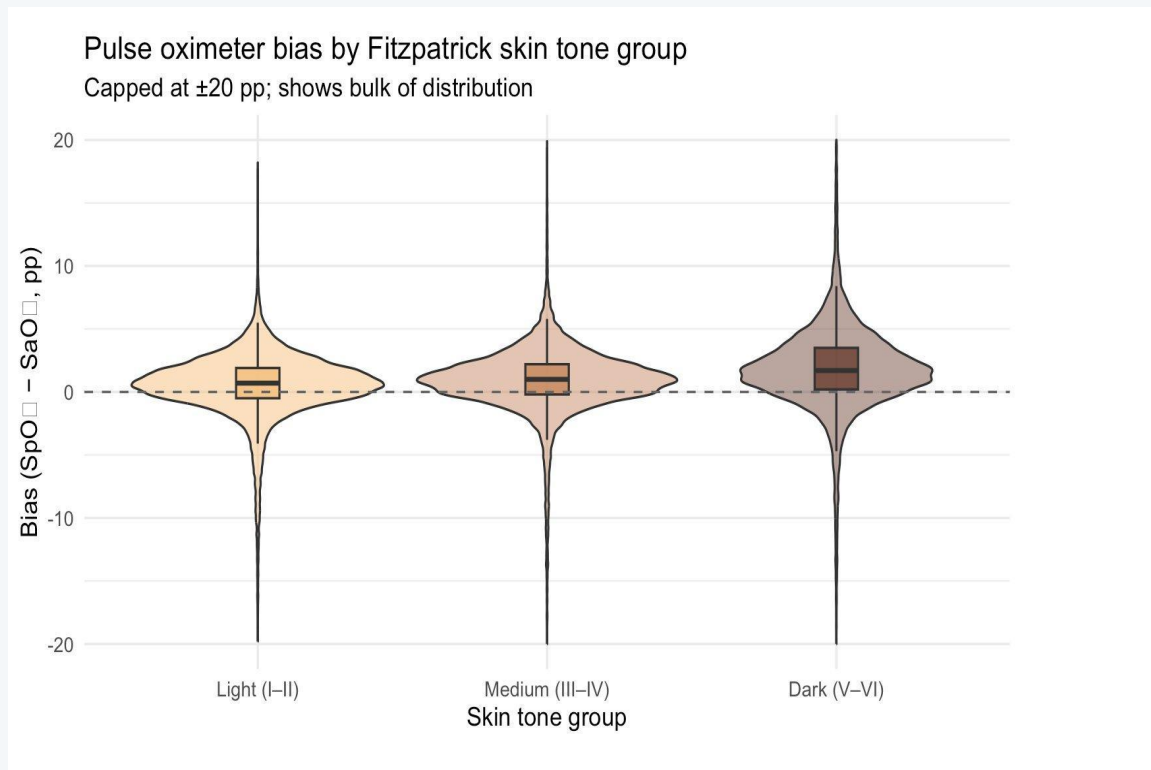
DATASET 2 · NY SPARCS

NY Dept of Health · 2021 Inpatient Discharges

- 103,907 respiratory discharge records (API — no download needed)
- Race, ethnicity, gender, age group
- Total charges, total costs, length of stay
- Insurance type, APR severity score, mortality outcome
- Free / public domain — NY Health Data portal

The two datasets do not share patients — the connection is made at the group level by joining race-stratified summaries from each.

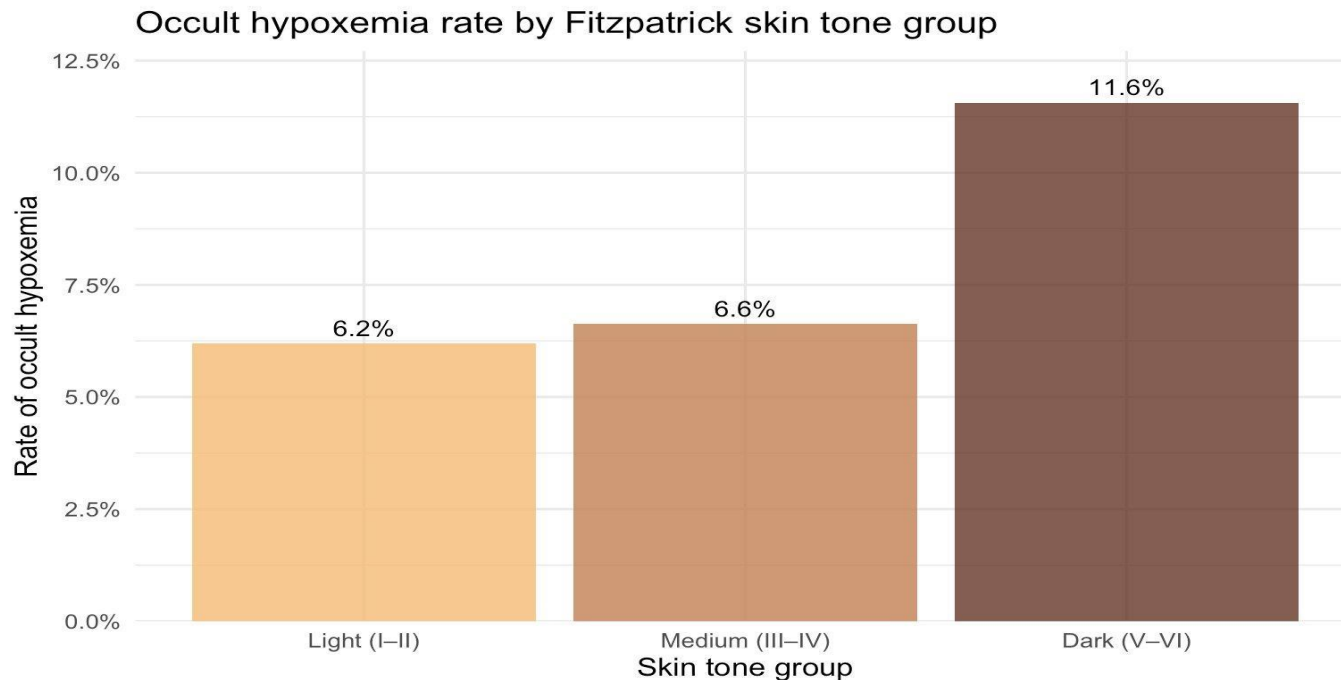
Pulse Oximeter Bias Increases with Skin Tone



What this means

- The bulk of readings cluster near zero for all groups — but the right tail extends further for darker skin.
- Even a 1–2 pp overestimate shifts clinical decisions: patients near the treatment cutoff may be denied oxygen.
- The bias is a gradient across all Fitzpatrick scores, not a threshold between two groups.
- The device flatters the reading for everyone, but darker-skinned patients are flattered most.

Dark Skin Tone -> Nearly Double the Rate of Hidden Hypoxemia

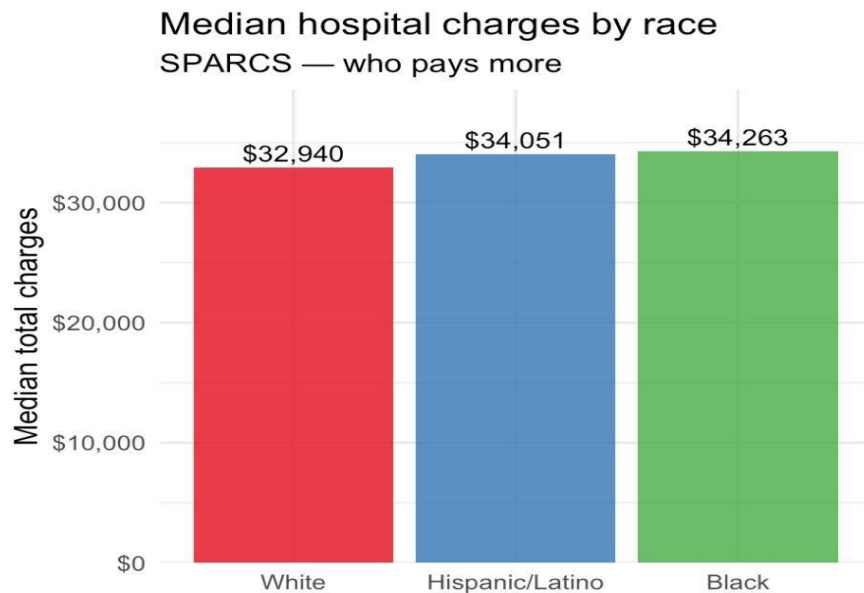
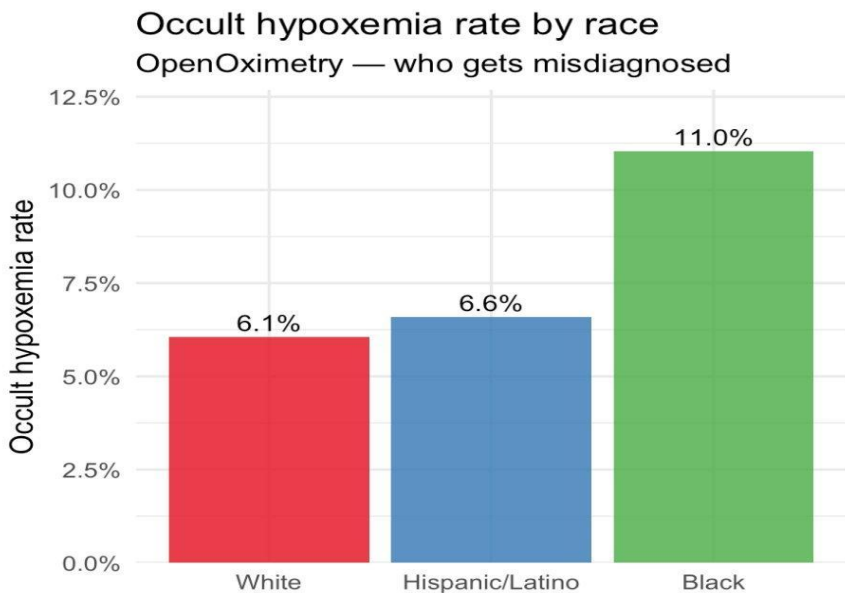


Patients with Fitzpatrick V–VI have an 11.6% occult hypoxemia rate vs. 6.2% for Fitzpatrick I–II — nearly double. This disparity exists regardless of actual oxygen need.

The Racial Ordering in Misdiagnosis Matches the Ordering in Costs

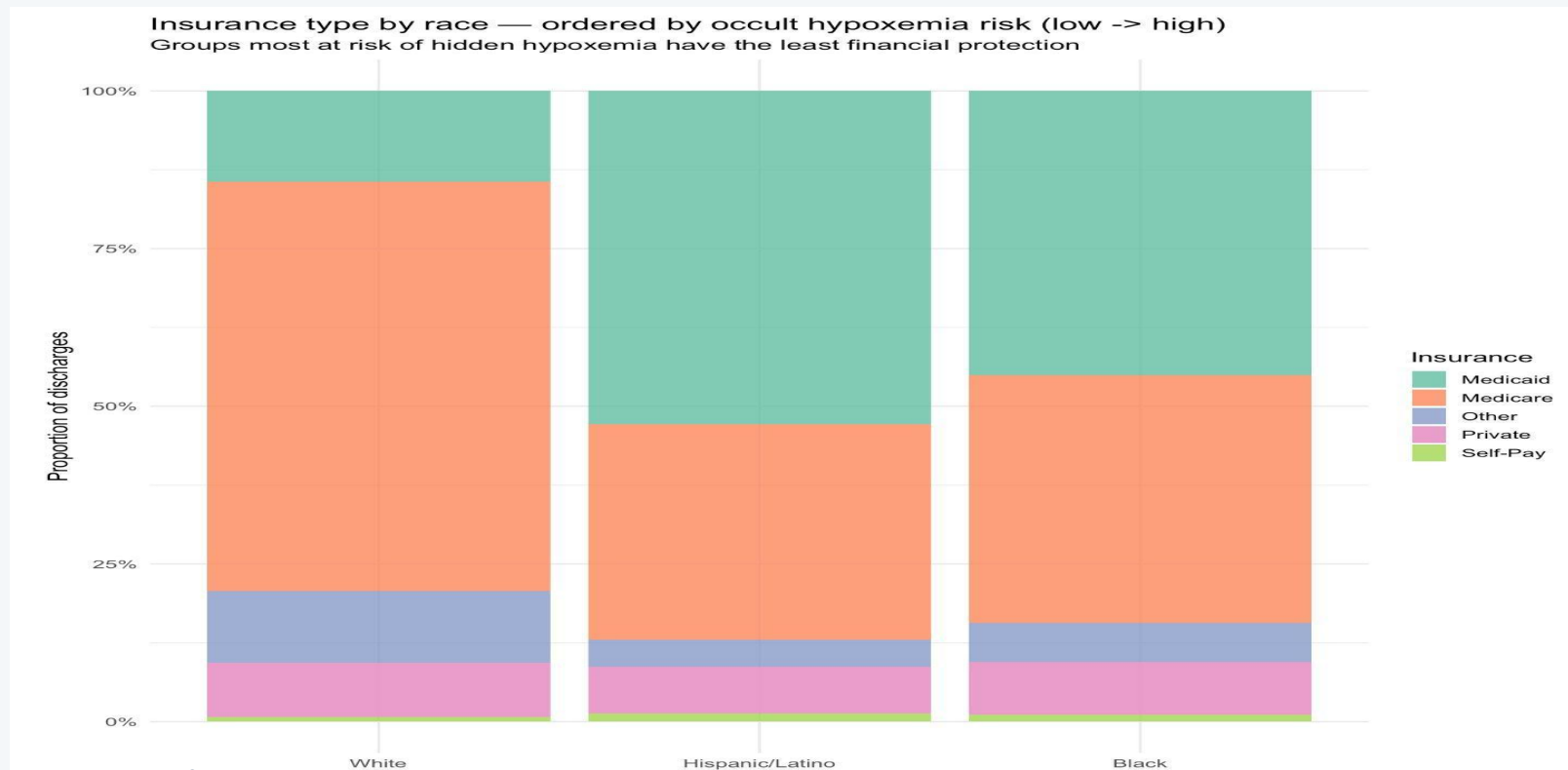
The same racial ordering appears in both misdiagnosis rates and hospital charges

Black patients: highest occult hypoxemia rate and highest median charges



Different populations, different cities — yet the direction is consistent: Black patients face the highest misdiagnosis rate and the highest charges.

Groups Most at Risk of Hidden Hypoxemia Have the Least Financial Protection

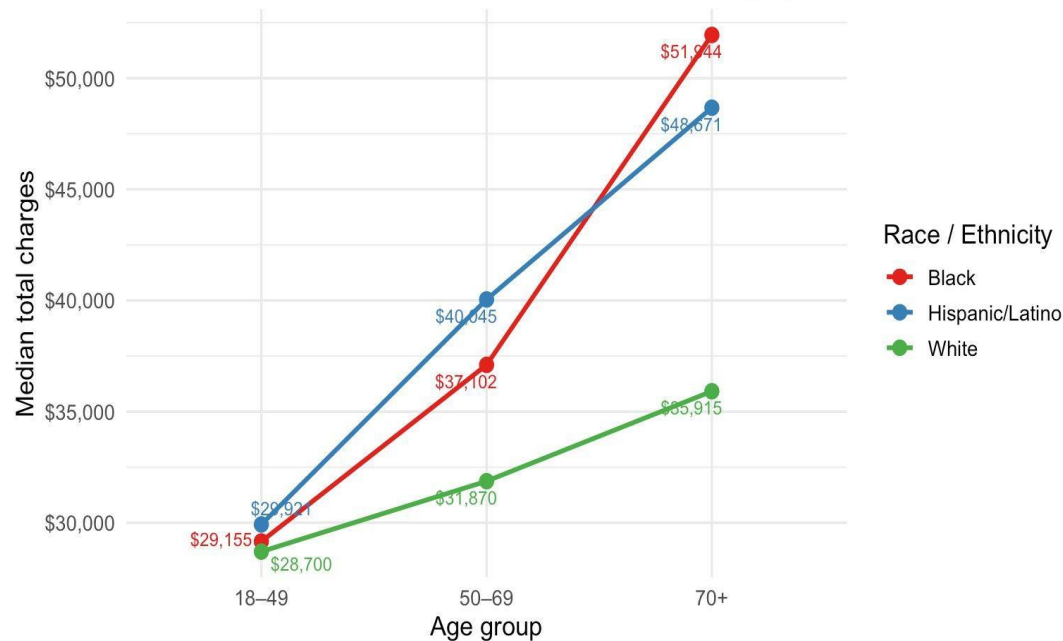


Moving left to right = increasing occult hypoxemia risk. Medicaid share rises as hypoxemia risk rises — the two vulnerabilities compound.

Black and Hispanic/Latino Patients Pay More at Every Age

Median charges by age group and race

Black and Hispanic/Latino patients are billed more at every age band



BY AGE BAND

18-49

Gap is small (~\$1500). All groups relatively healthy at this age.

50-69

Gap opens. Hispanic/Latino people ~\$8K more, Black people ~\$5K more than White people.

70+

Gap is largest. Black people pays ~\$16K more

At 70+, It is clear that undiagnosed patients are being charged more, and who pays that? Us! This issue is easily solvable if we paid more attention

INSURANCE CONVERGES

1% / 4% / 10%

White / Black / Hispanic on Medicaid
at 70+

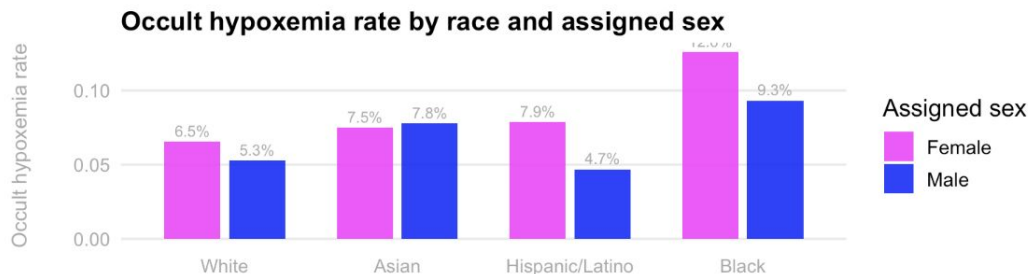
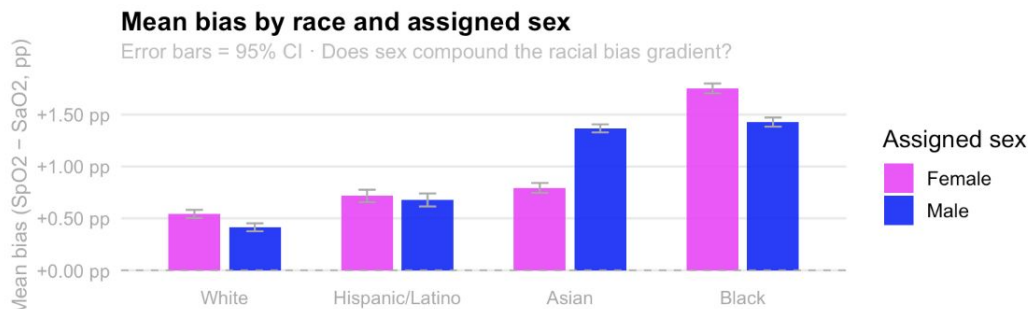
CHARGE GAP WIDENS

\$16K more

Black patients billed above White
patients at 70+
- after holding all other variables in
our datasets constant

Pulse Oximeter Bias is Prevalent in Individuals Assigned-Female at Birth

Sex × race interaction in pulse oximeter error



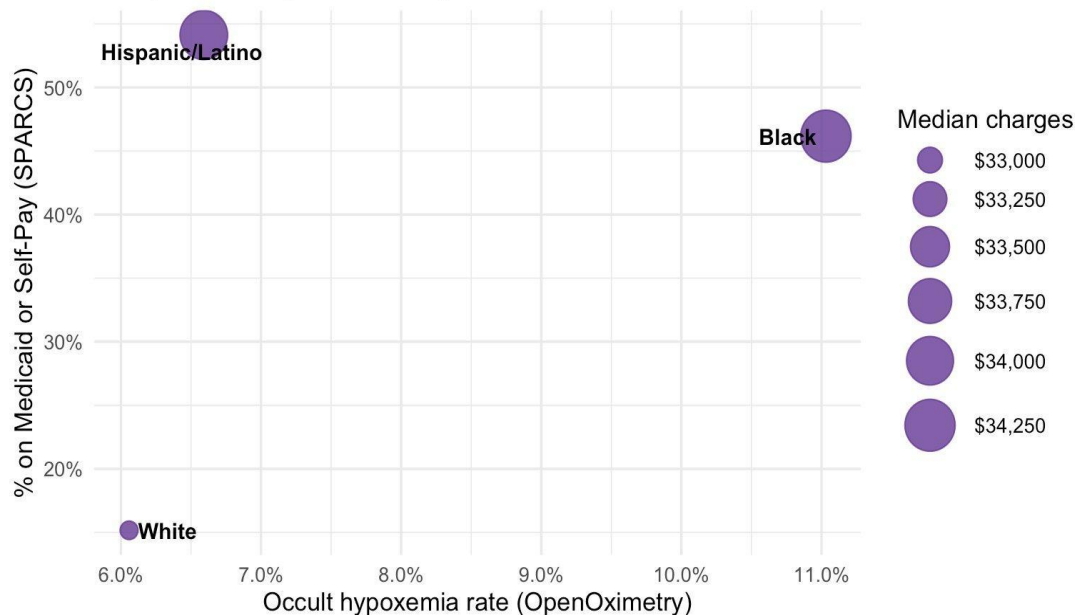
What this means

- While racial bias is a primary driver of device failure, being **female** often compounds this risk, particularly for **Black, Hispanic, Asian, and even White** patients, potentially leading to systemic delays in care for these specific groups.

The Double Burden: Who Is Most Exposed?

The double burden: hidden hypoxemia risk and financial vulnerability

Groups most likely to be misdiagnosed are also least insured



Black patients

Both elevated hypoxemia risk (~11%) and significant Medicaid exposure (~45%). Both vulnerabilities compound.

Hispanic/Latino

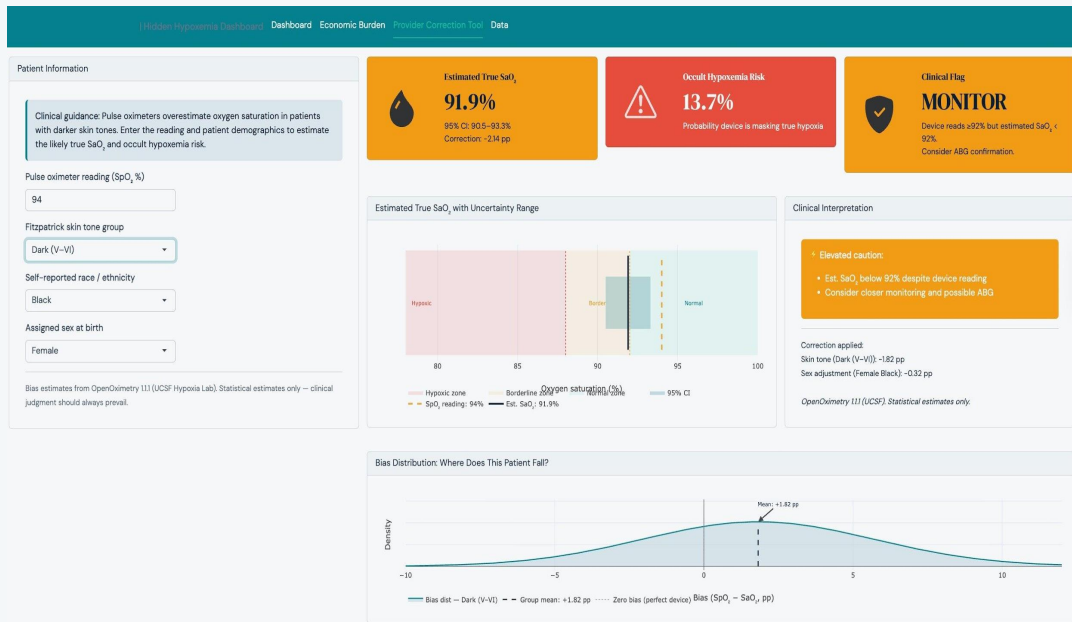
Extreme financial vulnerability (~55% Medicaid) with moderate hypoxemia risk (~6.6%) — additional drivers beyond device bias.

White patients

Lower on both dimensions; predominantly Medicare or Private insurance.

Live Bias Correction Tool for Clinical Use

Shiny App



How it works

Enter the pulse oximeter reading, Fitzpatrick skin tone group, race, and sex. The tool applies empirical bias corrections from OpenOximetry 1.1.1 (UCSF) and returns an estimated true SaO_2 with a 95% confidence interval.

Clinical flags

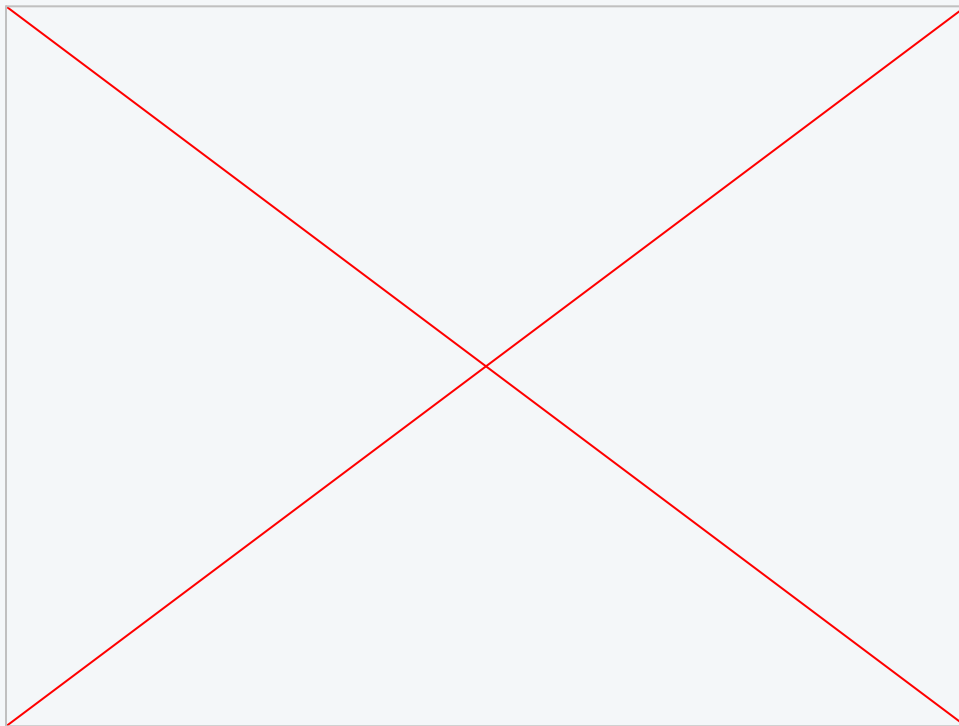
Three alert levels — MASKED (device reads $\geq 88\%$ but est. $\text{SaO}_2 < 88\%$), MONITOR ($\geq 92\%$ but est. $< 92\%$), and WITHIN RANGE — prompt the provider to consider ABG confirmation or withhold discharge.

What it shows

A gauge plot shows device reading vs. corrected estimate against the 88% and 92% thresholds. A bias distribution curve contextualizes the correction within the group-level error profile from the lab data.

Demo- Live Bias Correction Tool for Clinical Use

Shiny App



How the Bias Correction Tool Works

Bias
Correction

Measure the gap

1

Using 5,000+ paired readings from OpenOximetry, we know exactly how much each skin tone group's device reading differs from true blood oxygen on average.

Subtract the bias

2

$\text{Estimated SaO}_2 = \text{SpO}_2 - \text{Mean bias} - \text{Sex adj.}$

Example — Black female, Dark skin, reads 94%:
 $94 - 1.82 - 0.32 = 91.9\%$ true SaO₂

Flag the risk

3

If the corrected estimate falls below 88% or 92% — clinical treatment thresholds — the tool raises an alert so the provider can order a blood draw confirmation.

Bias parameters from OpenOximetry (n = 5,421 readings)

Skin group	Mean bias	Occlt rate	Std dev
 Light skin (I–II)	+0.62 pp	6.2%	±3.21
 Medium skin (III–IV)	+1.08 pp	6.6%	±3.45
 Dark skin (V–VI)	+1.82 pp	11.6%	±3.89

Clinical Flags

MASKED

Corrected est. < 88% → Patient may need O₂

MONITOR

Corrected est. < 92% → Consider blood draw

WITHIN RANGE

Corrected est. ≥ 92% → Continue monitoring

Formula: **Estimated SaO₂** = Device SpO₂ - **Group mean bias** - Sex adjustment | 95% CI = ±1.96 × SD / √30

Consequences for Patients, Providers, and Payers

CLINICAL IMPACT

- Delayed oxygen therapy increases risk of organ damage and mortality
- Patients denied treatment eligibility based on a flawed device reading
- Black patients show higher in-hospital mortality

ECONOMIC IMPACT ON PATIENTS

- Black patients billed ~\$16K more than White at 70+ for equivalent care
- 70% of Hispanic/Latino and 64% of Black working-age patients are on Medicaid — least protected when costs are highest
- Longer stays → higher out-of-pocket exposure for underinsured patients

IMPACT ON PROVIDERS & PAYERS

- Delayed treatment converts manageable cases into costly interventions
- Medicaid reimbursement shortfalls compound when stays are longer
- Systemic device bias creates legal and regulatory liability for health systems

Limitations

01**We are joining datasets based on race**

The two datasets do not share patients. Race-level summaries are joined across two different populations. Directional consistency is evidence — not proof of causation.

02**No Asian Patients category in SPARCS**

Asian patients are absorbed into 'Other Race' with no way to separate them, limiting the bridge to three racial groups.

03**SPARCS is New York only**

New York's insurance landscape may not generalize to states with different demographics and payer mixes.

Next Steps & Future Work

Geographic variation

Use hospital_county in SPARCS to test whether disparities are concentrated in specific regions or distributed across NY State.

Monk scale analysis

Test whether Monk skin tone scores, where available, capture additional variance in bias beyond the Fitzpatrick score.

Policy implications

Quantify what it would take for FDA oximeter standards to require validation across all Fitzpatrick scores before device clearance.

Thank You

The question this project asks is simple:

*When a medical device fails one group of patients more than another,
who pays — clinically and economically?*

GitHub: [mac-stat212-s26 / project-healthcare-william-patricia-nayla](#)

Site: [mac-stat212-s26.github.io/project-healthcare-william-patricia-nayla](#)

Data: OpenOximetry (PhysioNet) · NY SPARCS (NY Health Data)

Questions & Feedback Welcome